How Much Evidence Is Enough? A Bayesian Rare-Event Analysis of Driverless-Vehicle Crash Data

Independent reconstruction of Waymo's rider-only safety record from NHTSA Standing General Order reports, comparison with Waymo's published methodology, and a miles-to-demonstrate framework for any operator

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Executive summary

This report rebuilds the crash record of driverless (rider-only, RO) robotaxis from the raw crash reports filed with the U.S. National Highway Traffic Safety Administration (NHTSA) under Standing General Order 2021-01 (SGO). It analyses that record with the rare-event Bayesian methods of nuclear probabilistic risk assessment (PRA), then compares the approach, step by step, with the methodology Waymo publishes on its Safety Impact hub and in its peer-reviewed papers (Kusano et al., 2024, 2025; Scanlon et al., 2024; Chen et al., 2024). Every number below is traced to a figure or table in the body.

1. An independent pipeline reproduces Waymo's crash list almost exactly. Starting from 3,890 public SGO report rows, the pipeline isolates 1,934 Waymo rider-only crashed vehicles through June 2026. Of these, 1,925 (99.5%) appear in Waymo's own published crash list (Section 5.2, Figure 4, Tables 6–7). The outcome counts differ in ways consistent with Waymo's documented manual review. Waymo flags 20 more injury vehicles (17 of them have SGO severity "Unknown") and 9 more airbag deployments. Its serious-injury set excludes 3 SGO "Serious" cases, consistent with its police-report confirmation step.

2. Waymo's rates are well below the human benchmark, and the two inference methods agree. On our counts, the posterior median reduction is 89% for serious-or-worse injuries, 84% for airbag deployments and 84% for any reported injury (Figures 5–7, Tables 8–9). Applied to Waymo's verified counts, the Bayesian model gives the same central estimates as Waymo's published method, within 0.3 percentage points (94.5% against 94.8%, 81.7% against 81.7%, and 82.1% against 82.1%; Figure 7, Table 9). The differences between this analysis and Waymo's therefore come from data curation and benchmark choice, not from the statistics.

3. The probability of a large benefit is high, and it can be stated directly. There is a 95% probability that the true reduction exceeds 80% on every outcome (Figure 8, Table 10). The probability that it exceeds 75% is 0.993 even for serious injuries. A frequentist one-sided bound gives a similar number, but only a Bayesian analysis gives a direct probability statement about the reduction itself.

4. The largest unquantified uncertainty is in the benchmark corrections, not in the crash counts. Waymo publishes four versions of the any-injury human benchmark (observed or corrected for under-reporting, with or without spatial adjustment). They give reductions of 68.5% to 82.1% (Figure 10, Table 12), a spread more than twice the width of the headline 95% interval (78.9–85.0%). The corrections are well motivated, but they are applied as fixed values, and their own uncertainty is not carried into the headline interval. A bounding analysis across the four versions gives 65–84%. The conclusion of a large benefit is robust; the headline precision is likely overstated.

5. With zero events, a 100% point estimate is uninformative and a posterior is more useful. For city-level serious-injury results with zero events, Waymo's method reports a point estimate of "100% reduction" with intervals reaching down to −87% (Atlanta). The Bayesian median of 88%, with a probability of benefit of 0.95 under the Jeffreys prior (0.86 under a flat prior), summarizes the same evidence more usefully, while making the prior's influence explicit (Figure 11, Table 13).

6. Serious-injury claims are exposure-limited. A new operator needs about 9.1 million miles with zero serious crashes to claim, at 95% confidence, that it is better than humans. If its true

reduction is 80%, it reaches 80% power only at about 18.5 million miles; at a true 50% reduction it needs 82.5 million (Figures 13–14, Tables 15–16).

7. Being struck dominates the initiating events, and the crash partner sets severity. 62% of crashes occur while the Waymo vehicle is stationary, and these produce 0.36 of the 0.62 injury crashes per million miles (Figure 15, Table 17). Crashes with a cyclist or pedestrian injure someone 48% of the time, against 10% for light vehicles (Figure 16, Table 18).

8. Remote human operators are an open human reliability analysis (HRA) gap. In 5 of Tesla's 36 SGO narratives, a remote-assistance operator was controlling the vehicle at or just before contact (Table 20). We found no explicit model of these human actions in Waymo's published retrospective methodology.

1. Introduction and motivation

Driverless vehicles fail rarely, and most of their failures are minor. Safety claims are therefore a statistical problem before they are an engineering one. Kalra and Paddock (2016) showed that demonstrating fatality-rate superiority over human drivers could take hundreds of millions of miles. Operators, regulators and insurers need a framework that does three things:

1. turns sparse crash counts into calibrated uncertainty statements,
2. makes the dependence of those statements on exposure explicit, and
3. breaks crashes down into mechanisms that engineering can act on.

Nuclear PRA has faced this problem for decades. Initiating-event frequencies and component failure rates are estimated from a handful of events over large exposures, using Poisson likelihoods with Gamma priors, Beta-binomial branch probabilities, event trees and explicit treatment of epistemic uncertainty (Atwood et al., 2003).

Waymo is the leading publisher of retrospective AV safety data. Its methodology, described in Section 2.3 and compared in Section 6, is frequentist and carefully curated. This report applies the PRA toolkit to the same public data. It reproduces Waymo's results independently and identifies where a Bayesian, uncertainty-propagating treatment adds information. Figure 1 summarizes the workflow.

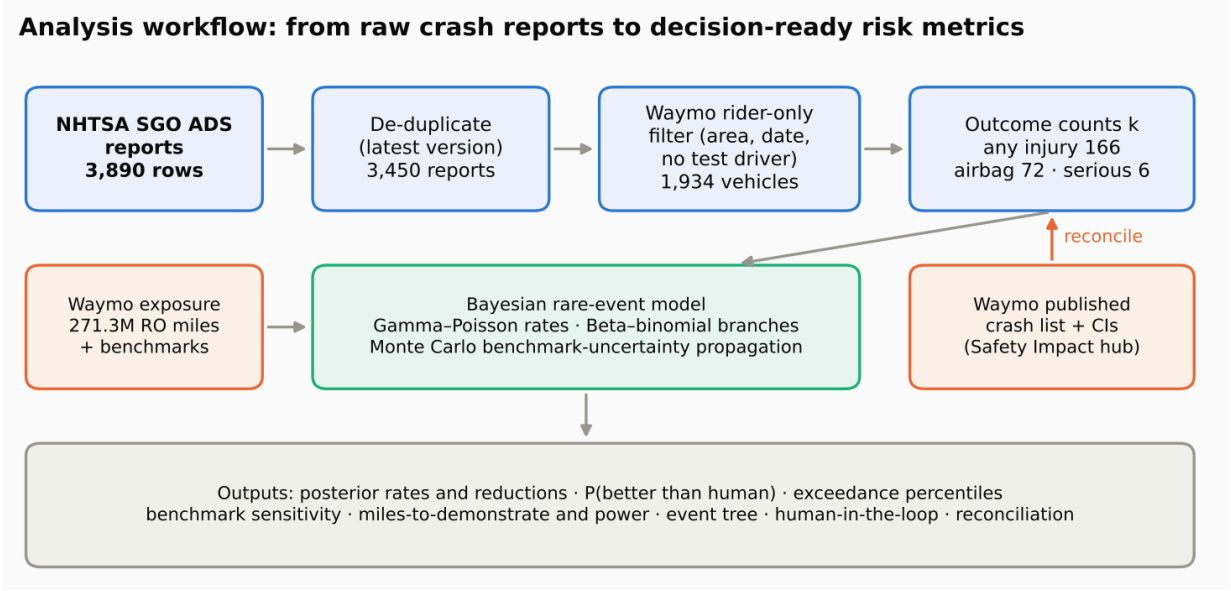


Figure 1. Analysis workflow. Raw SGO reports are de-duplicated, filtered to Waymo rider-only crashes and counted by outcome. The counts are combined with Waymo's published exposure and human benchmarks in a Bayesian rare-event model and reconciled against Waymo's own published crash list.

Figure 1 reads top to bottom. The top row is the data-processing chain (Section 3), which ends in the outcome counts k. The middle row brings in the two external inputs, Waymo's exposure and benchmarks (Table 2) and Waymo's published crash list. The orange "reconcile" arrow marks the independent check against Waymo's list (Section 5.2). The bottom box lists the outputs, each of which is a section of the results.

2. Data sources

2.1 NHTSA Standing General Order 2021-01 (ADS reports)

Since July 2021, SGO 2021-01 has required manufacturers and operators of vehicles with Automated Driving Systems (ADS) to report crashes that meet defined criteria. Each report describes one subject vehicle in one incident and may be updated in later versions. Table 1 lists the two public files used.

Table 1. NHTSA SGO ADS files used. The two files share no Report IDs.

File	Coverage (incident month)	Rows	Columns
Archive 2021–2025 `SGO-2021-01_Incident_Reports_ADS.csv`	Jul 2021 – Jun 2025	2,295	137
Current `SGO-2021-01_Incident_Reports_ADS.csv`	Apr 2025 – Sep 2026	1,595	116

The current file follows the Third Amended SGO (issued April 2025), which changed several fields and reporting thresholds. The fields used are Report ID and version, reporting entity, incident date, city and state, driver or operator type, engagement status, highest injury severity alleged, airbag deployment, crash partner, pre-crash movement and narrative. Coordinates and ZIP codes are redacted in most rows.

2.2 Exposure and human benchmarks

Only Waymo publishes rider-only mileage at the granularity rates require. Its Safety Impact hub (data through June 2026, release notes dated 2 October 2026) provides the exposure and city-level human benchmarks in Table 2. These are the inputs to every rate estimate in Sections 5.3–5.7.

Table 2. Waymo rider-only exposure and Waymo's headline human benchmarks (under-reporting corrected where applicable, spatially adjusted), through June 2026. City values are from the hub's downloadable comparison file; the all-location values are the rounded headline figures used in the aggregate analysis. IPMM = incidents (crashed vehicles) per million miles.

Metro area	RO miles (M)	Any-injury benchmark (IPMM)	Airbag, any vehicle (IPMM)	Serious or worse (IPMM)
Phoenix (Maricopa)	92.121	1.95	1.32	0.10
San Francisco Bay Area	82.421	6.64	1.92	0.39
Los Angeles	67.098	2.55	1.27	0.16
Austin (Travis)	21.064	3.26	2.32	0.17
Atlanta area	8.624	6.58	2.83	0.23
All locations	**271.3**	**3.77**	**1.61**	**0.21**

2.3 Waymo's published methodology and data

Waymo's retrospective methodology is set out on the Safety Impact hub and in four papers by its safety-research team. Kusano et al. (2024) compared crash rates at 7.1 million miles. Kusano et al. (2025) compared them by crash type at 56.7 million miles. Scanlon et al. (2024) constructed the human benchmarks, and Chen et al. (2024) developed the spatial ("dynamic") benchmark adjustment. A separate insurance-claims study by Swiss Re with Waymo (Di Lillo et al., 2024) found 88% fewer property-damage claims and 92% fewer bodily-injury claims than the overall driving population over 25.3 million miles. At a higher level, Waymo's safety case (Webb et al., 2020; Favarò et al., 2023) rests on hazard analysis (fault tree analysis, STPA, FMEA) and a "case credibility assessment". Retrospective crash-rate comparisons feed that case as evidence.

The essential features of Waymo's crash-rate methodology are as follows:

- **Crash selection.** Rider-only SGO crashes where the Waymo vehicle was in transport and was contacted.
  - Any injury: SGO severity of minor or above, plus "Unknown" severity cases whose narrative mentions an injury.
  - Airbag: SGO fields, checked against video review.
  - Serious or worse: SGO "Serious" or "Fatality" confirmed as A or K injuries on police reports obtained through public-records requests.
- **Benchmarks.** Built from state police-reported crash records and Highway Performance Monitoring System (HPMS) vehicle miles travelled, as crashed-vehicle rates on surface streets. The any-injury benchmark includes a 32% under-reporting correction (Blincoe et al., 2023). Benchmarks are re-weighted to Waymo's spatial driving distribution using S2 cells (Chen et al., 2024).
- **Inference.** Exact Poisson intervals for rates and Clopper–Pearson intervals for the ratio of two Poisson rates (Nelson, 1970). A result is called significant when the 95% interval of the percent

difference excludes zero. The papers state that no multiple-comparison correction is applied, and that under-reporting adjustments are treated as point estimates.

- **Published data.** The hub publishes the crash list with SGO IDs and outcome flags, the mileage by county, and the counts, rates and intervals for every outcome, location and crash type. This report downloaded those files and uses them for the reconciliation in Section 5.2 and the comparisons in Sections 5.3–5.6.

3. Data processing

3.1 Merging and de-duplication

The two NHTSA files were concatenated and their differing field names harmonized (Table 3). For each Report ID only the highest version was kept, with ties broken by the latest submission date, and placeholder rows were removed. This leaves **3,450 unique reports** from 51 reporting entities. Because each report describes one vehicle, counts in this report are *crashed vehicles*, the unit Waymo uses.

3.2 Outcome definitions

Table 3. Outcome definitions in this analysis and in Waymo's methodology. This analysis uses only fields any member of the public can download. Waymo adds manual review steps that require internal data.

Outcome	Archive file rule	Current file rule	Waymo's rule (for comparison)
Any injury	Severity ∈ {Minor, Moderate, Serious, Fatality}	Severity contains Minor, Moderate, Serious or Fatality	Same, plus "Unknown" with an injury in the narrative
Serious or worse	Severity ∈ {Serious, Fatality}	Severity contains Serious or Fatality	SGO Serious or Fatality confirmed as A or K on the police report
Airbag, any vehicle	SV airbag = Yes or crash-partner airbag = Yes	"Any Air Bags Deployed?" contains "Yes"	SGO fields, plus video review
Stationary SV	Pre-crash movement ∈ {Stopped, Parked}	same	not used

3.3 Isolating Waymo rider-only crashes

Waymo's exposure covers rider-only miles in five metro areas through June 2026, so the crash set must match that scope. Table 4 shows each filtering step and how many Waymo reports survive it.

Table 4. Selection funnel for the Waymo rider-only crash set.

Step	Rule	Waymo reports remaining
0	All Waymo reports after de-duplication	2,399
1	Incident month ≤ June 2026 (end of Waymo's exposure window)	2,221
2	In benchmarked areas: Arizona, California, and the Austin and Atlanta metros	2,152
3	Remove reports whose narrative mentions a test driver (214), whose engagement status or narrative shows the ADS not engaged (76), or that mention manual mode or takeover (89). Flags overlap.	**1,934**

California reports were split between the San Francisco Bay Area and Los Angeles by city name, because coordinates are redacted. This gives San Francisco Bay Area 822, Los Angeles 439, Phoenix 447, Austin 163 and Atlanta 62.

3.4 Narrative classification

The narratives were searched (case-insensitive) for indicators of human involvement:

- *Test driver present:* "test driver", "safety driver", "vehicle operator was", "in the driver's seat".
- *Remote operator in control:* "remote support / assistance / operator / driver / control", "remotely took control", "teleoperat".
- *Manual or takeover:* "manual mode", "disengaged", "took over", "took control".
- *Any human role:* any of the above, or "safety monitor".

Keyword rules are transparent but approximate, and results derived from them are labelled as such.

4. Statistical methodology

Figure 2 illustrates the core idea before the equations: the posterior distribution of a crash rate combines a deliberately weak prior with the evidence in the count and the exposure.

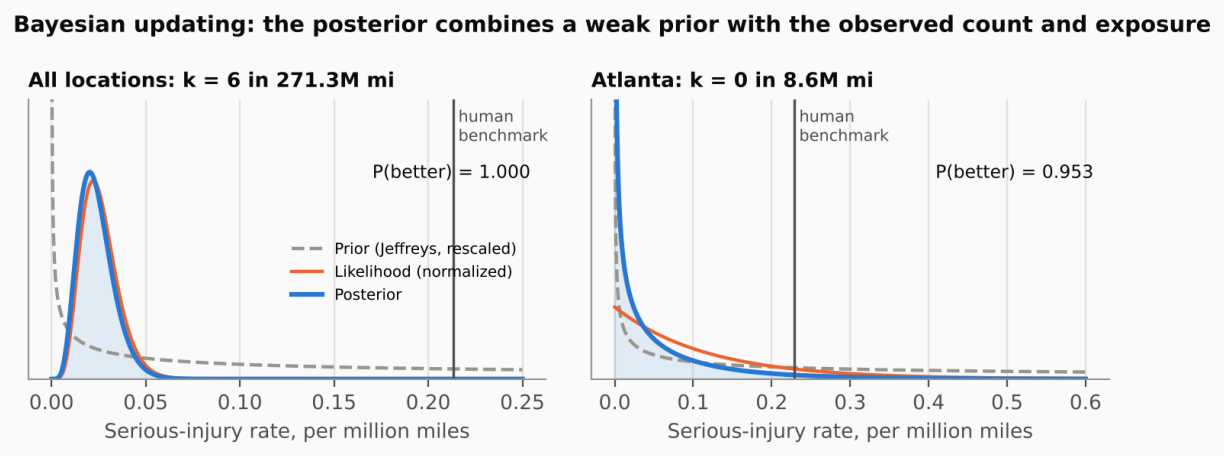


Figure 2. Bayesian updating for the serious-injury rate. Left: all locations, k = 6 events in 271.3 million miles. Right: Atlanta, k = 0 in 8.6 million miles. Gray dashed: Jeffreys prior (rescaled for display). Orange: likelihood. Blue: posterior. The shaded area is the posterior probability that the rate is below the human benchmark (vertical line).

In the left panel of Figure 2, the evidence dominates. The posterior (blue) almost coincides with the likelihood (orange), and essentially all of its mass lies below the benchmark of 0.21, so P(better) ≈ 1.000. In the right panel, Atlanta has zero serious crashes but little exposure. The posterior is still informative, since most mass sits below 0.10, but its long right tail crosses the benchmark of 0.23, so P(better) = 0.953. The frequentist approach (Section 6) reports this same situation as a point estimate of “100% reduction”.

4.1 Likelihood

For an outcome, let *k* be the number of crashed vehicles observed over exposure *M* (million miles). Crashes are modelled as a Poisson process with rate λ per million miles:

$$P(\,k \mid \lambda, M\,) \;=\; \frac{(\,\lambda M\,)^{\,k} e^{-\lambda M}}{k!}$$

This is the model PRA uses for initiating-event frequencies. Its main assumption, a constant rate that is independent across miles, is discussed in Section 8.

4.2 Prior and posterior

A Gamma prior $\lambda \sim \mathrm{Gamma}\left(\,\alpha_0, \beta_0\,\right)$ is conjugate to the Poisson likelihood:

$$p\,(\,\lambda \mid k, M\,) \;\propto \lambda^{k} e^{-\lambda M}\, \lambda^{\alpha_0 - 1} e^{-\beta_0 \lambda} \;\Rightarrow\; \lambda \mid k, M \sim \mathrm{Gamma}\left(\,\alpha_0 + k,\; \beta_0 + M\,\right)$$

The Jeffreys prior $p\,(\,\lambda\,) \;\propto \lambda^{-1/2}$ ($\alpha_0 = 1/2$, $\beta_0 \to 0$) is the default for Poisson rates in nuclear PRA data analysis (Atwood et al., 2003). It gives

$$\lambda \mid k, M \sim \mathrm{Gamma}\left(\,k + \tfrac{1}{2},\; M\,\right), \qquad E\left[\,\lambda \mid k, M\,\right] \;=\; \frac{k + 1/2}{M}$$

The prior contributes the equivalent of half an event and almost no exposure, so the data dominate (Figure 2, left). It still gives a proper, informative posterior when $k = 0$ (Figure 2, right).

4.3 Credible intervals, percentiles and the probability of benefit

Posterior percentiles are $F^{-1}(p)$ of the Gamma posterior. The 95% credible interval (CrI) uses $p = 0.025$ and 0.975, and the 50% interval uses 0.25 and 0.75. With λ_H as the human benchmark, the probability that the fleet is safer is

$$P(\lambda < \lambda_H \mid k, M) = \frac{\gamma\left(k + \frac{1}{2},\ M\lambda_H\right)}{\Gamma\left(k + \frac{1}{2}\right)}$$

where γ is the lower incomplete gamma function.

4.4 Relative reduction and exceedance probability

The quantity behind a safety claim is the relative reduction $R = 1 - \lambda/\lambda_H$. Its distribution was obtained from 200,000–400,000 Monte Carlo draws of λ. For decisions, the *exceedance curve*

$$E(x) = P(R > x \mid k, M), \qquad 0 \le x \le 100\%$$

gives the probability that the true reduction exceeds any chosen threshold (Section 5.3).

4.5 Uncertainty in the human benchmark

Two kinds of benchmark uncertainty are treated.

Parametric uncertainty (sampling, under-reporting, adjustment) is represented by a lognormal benchmark with its mean at the published value and coefficient of variation $c \in \{0, 0.2, 0.4\}$:

$$\lambda_H \sim \text{LogNormal}\left(\ln \bar{\lambda}_H - \frac{\sigma^2}{2},\ \sigma^2\right), \qquad \sigma^2 = \ln\left(1 + c^2\right)$$

Model-form uncertainty arises from the choice among Waymo's alternative benchmark constructions B_j. It is represented by an equal-weight mixture:

$$p(R \mid k, M) = \sum_{j=1}^{J} w_j\, p\left(R \mid k, M, B_j\right), \qquad w_j = 1/J$$

The separation between the Waymo posterior and the uncertain benchmark is summarized by the overlap coefficient

$$\text{OVL} = \int \min\left\{f_W(\lambda),\ f_H(\lambda)\right\} d\lambda$$

which is 0 for complete separation and 1 for identical distributions.

4.6 Miles needed to demonstrate a claim, and power

An operator observing *k* events can claim at confidence $1 - a$ (here 0.95) that its rate is below $f\lambda_H$ once its exposure reaches

$$M^*(k) = \frac{Q_{\text{Gamma}(k + 1/2,\ 1)}(1 - a)}{f\lambda_H}$$

where Q is the unit-rate Gamma quantile. For $k = 0$, $Q = 1.92$, which is less conservative than the classical "rule of three" ($Q = 3.00$, flat prior). Prospectively, for a fleet with true rate $\lambda_T = (1 - r)\,\lambda_H$,

the *power* to make the claim by exposure *M* is

$$\pi\left(M\right) = P\Big(K \le k^{*}\left(M\right)\Big), \quad K \sim \text{Poisson}\left(\lambda_T M\right), \quad k^{*}\left(M\right) = \max\left\{k : P\left(\lambda < \lambda_H \mid k, M\right) \ge 0.95\right\}$$

4.7 Severity branches as an event tree

Each crash is an initiating event that branches into end states. For branch *b* with n_b crashes and x_b injury crashes, the Jeffreys Beta posterior is

$$p_b \mid x_b \sim \text{Beta}\left(x_b + \tfrac{1}{2},\ n_b - x_b + \tfrac{1}{2}\right)$$

and the end-state frequency is

$$\lambda_{b,\,\text{inj}} = \lambda_{\text{crash}}\ P\left(b \mid \text{crash}\right)\ P\left(\text{injury} \mid b\right), \qquad \lambda_{\text{inj}} = \sum_{b} \lambda_{b,\,\text{inj}}$$

4.8 Waymo's frequentist method, reproduced for comparison

Waymo's interval for the rate ratio $\rho = \lambda_W / \lambda_H$ (Nelson, 1970) conditions on the total count. With $k_W \sim \text{Poisson}\left(\lambda_W M_W\right)$ and $k_H \sim \text{Poisson}\left(\lambda_H M_H\right)$, and $n = k_W + k_H$,

$$k_W \mid n \sim \text{Binomial}\left(n, \pi\right), \qquad \pi = \frac{\rho M_W}{\rho M_W + M_H}, \qquad \rho = \frac{\pi}{1-\pi} \cdot \frac{M_H}{M_W}$$

A Clopper–Pearson interval for π maps to an interval for ρ, and the percent reduction is $1 - \rho$. The point estimate is $\widehat{\rho} = \left(k_W / M_W\right) / \left(k_H / M_H\right)$, which equals 0 (a "100% reduction") whenever $k_W = 0$.

4.9 Computation

All calculations use Python (pandas, NumPy, SciPy, Matplotlib) in three scripts:

- `sgo_analysis.py`: data processing and the core model.
- `compare_waymo.py`: reconciliation with Waymo's crash list.
- `comparison_figs.py`: method comparison and the uncertainty figures.

Posterior quantiles are exact, and Monte Carlo draws use fixed seeds.

5. Results

5.1 Industry overview

Table 5. SGO ADS reports by operator after de-duplication (all operating modes and locations). General Motors co-reported many Cruise incidents, so those two rows overlap.

Operator	Reports	Any injury	Serious or fatal	Airbag (any)	SV stationary (%)	First – last incident
Waymo	2,399	213	11	105	58	2021-07 – 2026-09
Zoox	172	35	0	9	37	2021-07 – 2026-09
Cruise	155	26	3	12	32	2021-07 – 2024-11
General Motors	149	25	3	13	32	2021-08 – 2024-11
Transdev	140	12	0	5	46	2021-07 – 2025-05
Avride	84	3	0	8	21	2023-10 – 2026-08
May Mobility	48	4	0	1	33	2021-07 – 2026-08
Tesla	36	2	0	0	36	2025-07 – 2026-08

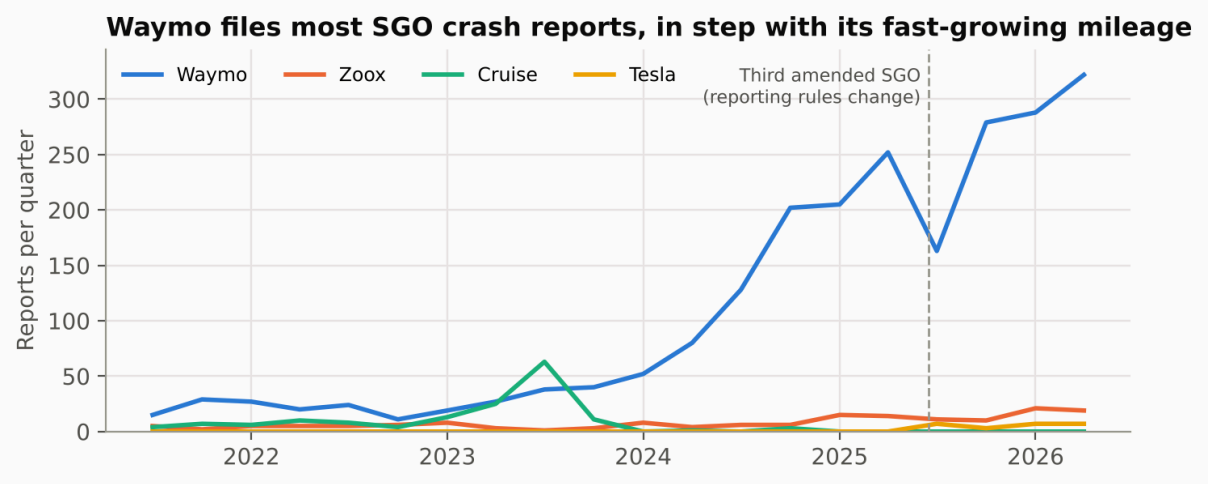


Figure 3. SGO crash reports per quarter for four operators. The dashed line marks the transition to the Third Amended SGO in mid-2025.

Table 5 and Figure 3 show that Waymo files about 70% of all ADS reports (2,399 of 3,450), and that its quarterly count grew from about 20 in 2022 to more than 300 in Q2 2026. The dip in Q3 2025 coincides with the reporting-rule change. A report count measures activity, not safety: without exposure, a rising count can mean either more miles or a higher crash rate. Rates are therefore estimated only for Waymo, the one operator that publishes rider-only miles.

5.2 Reconciliation with Waymo's published crash list

Waymo's published crash list was matched to this analysis' set by SGO Report ID. The list has 1,977 rows; 2 predate the SGO and have no ID, and 41 lie outside the five benchmarked metros.

Table 6. Crashed-vehicle sets: this analysis against Waymo's published crash list (SGO era, five benchmarked metros).

Set	Crashed vehicles
Waymo list, SGO era, five metros	1,929
This analysis	1,934
In both	1,925
Only in this analysis	9
Only in Waymo's list	4

Table 7. Outcome flags: this analysis against Waymo's published crash list, with the main reason for each difference.

Outcome	This analysis	Waymo (SGO era)	Both flag	Only this analysis	Only Waymo	Main reason for the difference
Any injury	166	182	162	4	20	17 of Waymo's 20 extra cases have SGO severity "Unknown" and an injury in the narrative
Airbag, any vehicle	72	80	71	1	9	Consistent with Waymo's video review of airbag deployments not recorded in SGO fields
Serious or worse	6	3	3	3	0	3 SGO "Serious" cases absent from Waymo's serious set, consistent with its police-report confirmation

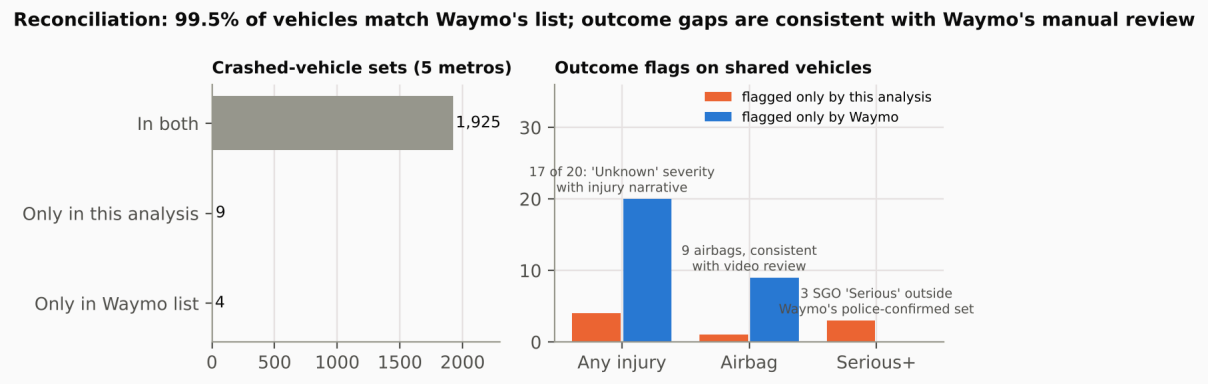


Figure 4. Reconciliation with Waymo's crash list. Left: vehicle sets. Right: outcome flags that only one side assigns, annotated with the reason.

Figure 4 and Tables 6–7 show that a fully automated pipeline using only public fields recovers 99.5% of Waymo's curated crash set. The 13 mismatched vehicles can be traced:

- **Waymo's 4 extra vehicles.** Our engagement filter removed one and our takeover filter another. One had its location redacted, so it failed the area filter. One was not in the public files downloaded for this analysis.
- **Our 9 extra vehicles.** Eight are 2021 crashes in San Francisco and Chandler. Their early-format narratives do not mention a test driver, so our filter kept them. They are absent from Waymo's rider-only list, and Waymo's rider-only operation in San Francisco began after 2021. These are most likely supervised-testing crashes that this analysis misclassified as rider-only. The ninth is a 2024 Austin crash.

The outcome differences reflect methodological choices rather than errors. Waymo's manual review adds information that is not in the public fields (narratives, video, police reports), so its injury and airbag counts are higher and its serious count is lower. Of Waymo's 20 extra injury vehicles, 12 are in Phoenix. Section 5.3 shows that these differences shift the estimated reductions by at most about 5 percentage points.

5.3 Aggregate posterior rates, percentiles and reductions

Table 8. Posterior rate percentiles for Waymo rider-only crashes (this analysis' counts, 271.3 million miles). The posterior probability that the rate is at or above the benchmark is below 10^{-17} for all three outcomes.

Outcome	*k*	Posterior median IPMM	50% interval	95% interval	Benchmark IPMM	P(better than human)
Serious injury or worse	6	0.023	0.017–0.029	0.009–0.046	0.21	> 0.9999
Airbag, any vehicle	72	0.266	0.245–0.288	0.209–0.332	1.61	> 0.9999
Any injury reported	166	0.612	0.581–0.645	0.524–0.710	3.77	> 0.9999

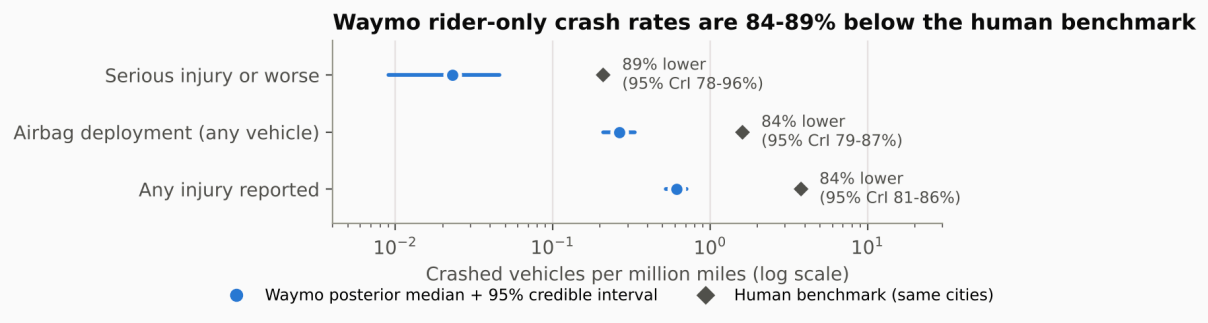


Figure 5. Posterior median and 95% credible interval for each outcome (blue) against the human benchmark (gray diamond), on a log scale.

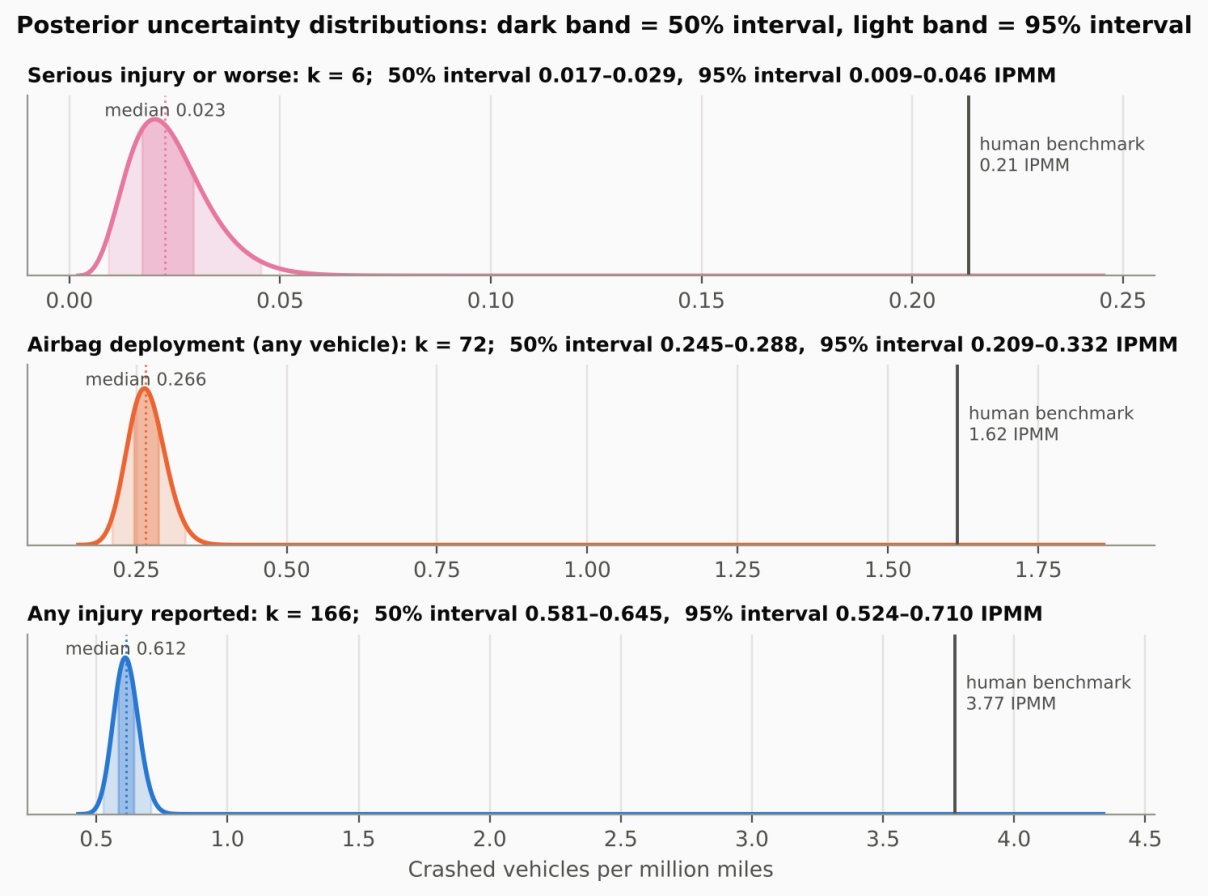


Figure 6. Posterior uncertainty distributions. The dark band is the 50% interval, the light band the 95% interval, the dotted line the median, and the solid vertical line the human benchmark.

Figure 5 summarizes the three posteriors on one log axis. Each 95% interval lies a factor of 5–10 below its benchmark. Figure 6 shows the full distributions and percentiles behind Table 8.

- **Serious injury** (top panel, six events) is the widest distribution. Its 95% interval spans a factor of five (0.009–0.046), and it is right-skewed, as small-count Gamma posteriors are.
- **Airbag and any-injury** rates (lower panels) are tight and nearly symmetric, because they rest on 72 and 166 events.
- In all three panels the benchmark line sits far outside even the light 95% band, which is why P(better) exceeds 0.9999.

Table 9. Reduction relative to the human benchmark under three approaches. "Bayesian with Waymo's counts" applies this report's model to Waymo's verified counts (183, 80 and 3 over 271.3 million miles).

Outcome	This analysis: median (95% CrI)	Bayesian with Waymo's counts: median (95% CrI)	Waymo published: point (95% CI)	P(reduction > 75%), this analysis
Serious injury or worse	89.3% (78.6–95.7%), $k = 6$	94.5% (86.1–98.5%), $k = 3$	94.8% (83.2–99.2%)	0.993
Airbag, any vehicle	83.5% (79.4–87.0%), $k = 72$	81.7% (77.4–85.4%), $k = 80$	81.7% (76.6–86.0%)	> 0.999
Any injury reported	83.8% (81.2–86.1%), $k = 166$	82.1% (79.4–84.6%), $k = 183$	82.1% (78.9–85.0%)	> 0.999

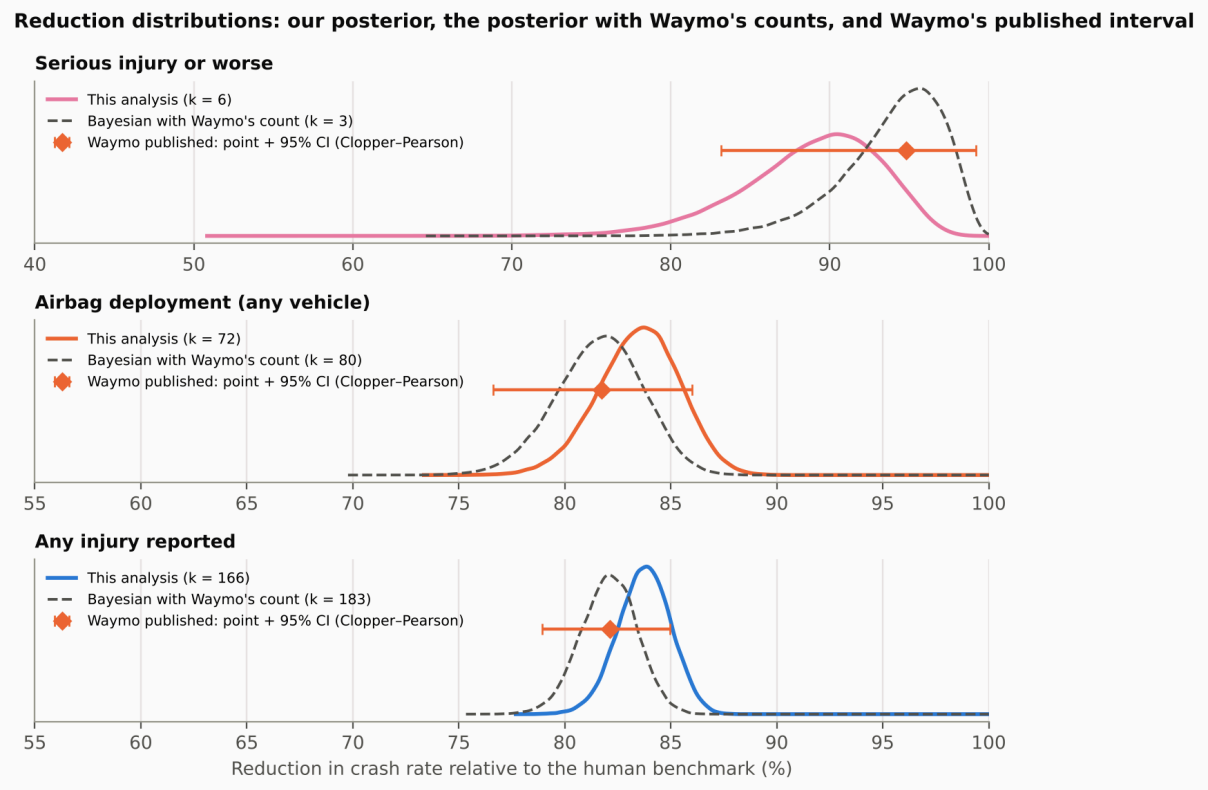


Figure 7. Reduction distributions. Solid: this analysis. Dashed: the same Bayesian model with Waymo's verified counts. Orange diamond and bar: Waymo's published point estimate and Clopper–Pearson 95% interval.

Figure 7 and Table 9 separate two sources of difference.

The data source is the gap between the solid and dashed curves. It reflects the counting differences of Section 5.2 and moves the medians by 1.7–5.2 points. Our higher serious-injury count (6 against 3) gives the more conservative 89.3%.

The inference method is the gap between the dashed curve and the orange marker. With identical counts, the Bayesian medians match Waymo's point estimates to within 0.3 points. The Bayesian 95% intervals are slightly narrower, because Clopper–Pearson intervals are conservative by construction (coverage ≥ 95%) and because this model treats the benchmark as fixed (Section 5.4 relaxes that).

The two methods therefore agree on the evidence; they differ in what they can say about it, which the next figure shows.

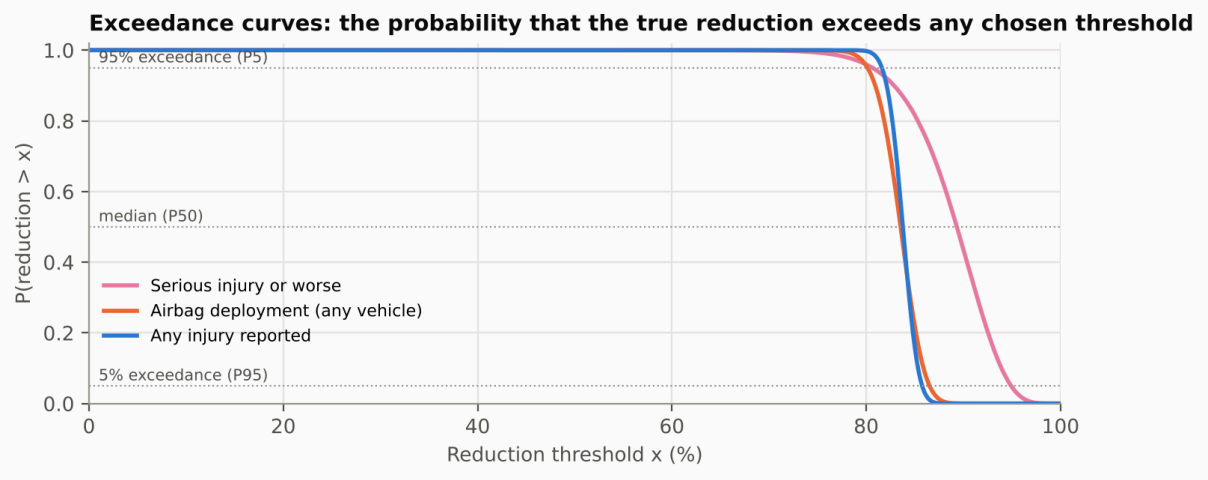


Figure 8. Exceedance curves: the probability that the true reduction exceeds threshold x, for each outcome.

Table 10. Exceedance percentiles of the reduction (this analysis' counts).

Outcome	95% probability the reduction is at least (P5)	Median (P50)	5% probability the reduction is at least (P95)
Serious injury or worse	80.6%	89.3%	94.9%
Airbag, any vehicle	80.1%	83.5%	86.5%
Any injury reported	81.6%	83.8%	85.8%

Figure 8 turns each reduction distribution into a decision curve. Read upward from a threshold on the x-axis: at x = 75% all three curves are near 1, so a "75% safer" claim is supported with probability above 0.99. The curves fall steeply between 80% and 87% for airbag and any-injury, because the evidence pins those reductions down. The serious-injury curve falls more gradually between 80% and 95%, the visual signature of only six events. Table 10 reads off three horizontal slices.

These are the statements a regulator, insurer or safety board usually wants ("there is a 95% probability that the reduction is at least 80%"). A frequentist one-sided 95% bound would give a similar number, but its interpretation is about the procedure, not a probability for the reduction itself.

5.4 Uncertainty in the human benchmark

Table 11. Median reduction, with its 2.5th-percentile lower bound in parentheses, as parametric benchmark uncertainty grows. The last column is the overlap coefficient between the Waymo posterior and the uncertain benchmark.

Outcome	CV = 0	CV = 0.2	CV = 0.4	Overlap coefficient at CV = 0.4
Serious injury or worse	89.2% (≥ 78.3%)	89.0% (≥ 75.4%)	88.5% (≥ 67.1%)	0.003
Airbag, any vehicle	83.5% (≥ 79.4%)	83.2% (≥ 73.6%)	82.3% (≥ 61.0%)	< 0.001
Any injury reported	83.8% (≥ 81.2%)	83.5% (≥ 74.9%)	82.5% (≥ 62.3%)	< 0.001



Figure 9. Overlap between the Waymo posterior (colored) and a human benchmark with 20% (dashed) and 40% (solid) coefficient of variation. Peaks are scaled to 1 so the shapes can be compared.

Figure 9 asks whether the Waymo distribution could be confused with the human one if the benchmark were much less certain than published. Even with a 40% coefficient of variation, the human distributions (gray) barely touch the Waymo posteriors: the overlap coefficient is at most 0.003 (Table 11). The medians in Table 11 hardly move as the benchmark becomes uncertain; only the lower tail widens, from about 80% to about 61–67%. Parametric benchmark uncertainty therefore cannot overturn the conclusion. The benchmark's *construction* is a different matter.

Table 12. Any-injury reduction under Waymo's four published benchmark constructions (Waymo's count, k = 183).

Benchmark construction (any injury, all locations)	**Benchmark IPMM**	**Waymo published reduction (95% CI)**	**Bayesian median (95% CrI)**
Observed, unadjusted	2.14	68.5% (62.9–73.5%)	68.5% (63.7–72.9%)
Observed, spatially adjusted	2.58	73.9% (69.2–78.0%)	73.8% (69.9–77.5%)
Under-reporting corrected, unadjusted	3.14	78.5% (74.6–81.9%)	78.5% (75.2–81.4%)
Under-reporting corrected, spatially adjusted (headline)	3.77	82.1% (78.9–85.0%)	82.1% (79.4–84.6%)
Equal-weight mixture of the four	—	—	**76.3% (65.4–83.8%)**

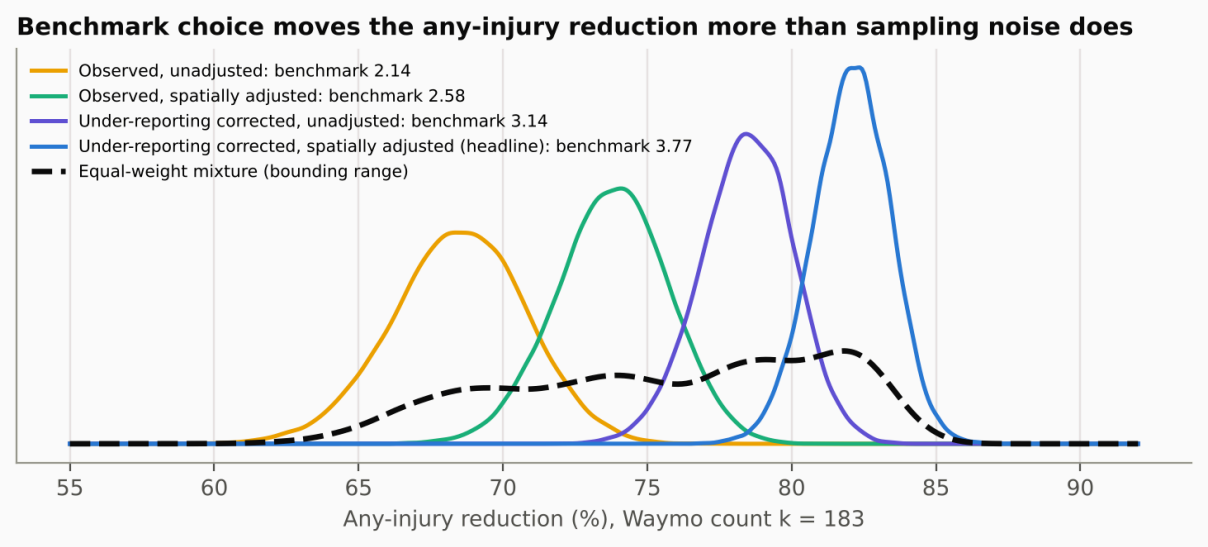


Figure 10. Benchmark model-form uncertainty. Colored curves: reduction posterior under each of Waymo's four benchmark constructions. Black dashed: the equal-weight mixture, a bounding range rather than a best estimate.

Figure 10 is the most consequential result of the comparison. Each colored curve is the reduction posterior under one of Waymo's own benchmark constructions, all using the same 183 crashes. Each curve's 95% interval is 5–9 points wide, which is the sampling uncertainty Waymo reports. The curves' centers, however, span 14 points (68.5% to 82.1%), because the 32% under-reporting correction and the spatial re-weighting each shift the benchmark substantially.

The four versions are not equally credible. Police-reported injury crashes are known to be under-reported, so the observed (uncorrected) benchmarks are biased low, and Waymo's choice of the fully corrected version as its headline is well motivated. The gap is different: the corrections themselves are uncertain. The 32% under-reporting share is a national estimate from 2019 data (Blincoe et al., 2023), and Chen et al. (2024) report bootstrap intervals for the spatial multipliers. Neither uncertainty is carried into the headline interval, whose benchmark bounds (3.74–3.80 IPMM) are too narrow to include correction uncertainty and appear to reflect sampling only.

The black mixture curve is therefore a *bounding* analysis, not a best estimate. It shows the range of conclusions supported by Waymo's own published constructions: 65–84%, against 79–85% for the headline alone (Table 12). The benefit remains large under every construction, with even the most conservative giving a reduction above 60%. The headline's precision, however, depends on correction factors whose uncertainty is not quantified. A full treatment would propagate distributions for the under-reporting share and the spatial multipliers (Section 9).

5.5 City-level results: frequentist and Bayesian intervals compared

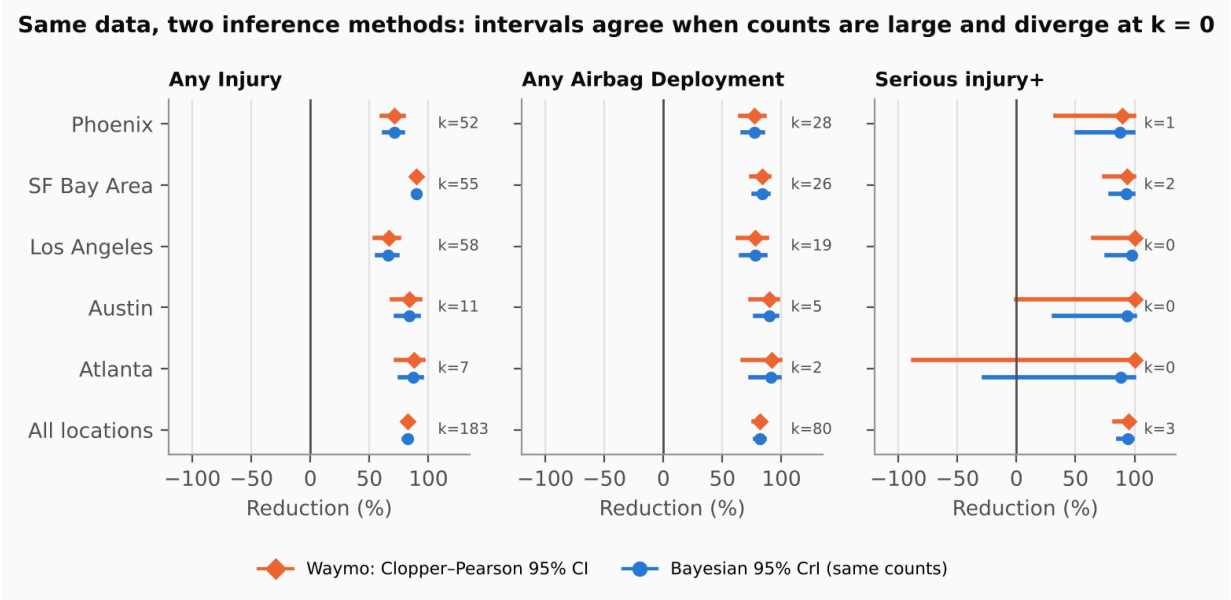


Figure 11. Waymo's Clopper–Pearson 95% intervals (orange) and Bayesian 95% credible intervals (blue) computed from the same counts, by outcome and metro area. The number on the right is the event count k.

Table 13. Serious-or-worse reduction by metro: Waymo's frequentist results against Bayesian results from the same counts.

Metro	Serious *k*	Miles (M)	Waymo published: point (95% CI)	Bayesian: median (95% CrI)	P(better)
Phoenix	1	92.1	89.5% (33.0–99.9%)	87.6% (51.1–98.9%)	0.9997
SF Bay Area	2	82.4	93.8% (74.6–99.5%)	93.3% (80.0–98.7%)	> 0.9999
Los Angeles	0	67.1	100% (64.8–100%)	97.8% (76.1–100%)	> 0.9999
Austin	0	21.1	100% (−0.2–100%)	93.8% (32.0–100%)	0.994
Atlanta	0	8.6	100% (−87.4–100%)	88.5% (−27.3–100%)	0.953

Figure 11 places the two methods side by side for every outcome and city. In the any-injury and airbag panels (left and center), with $k = 2$ to 183, the orange and blue intervals nearly coincide: when evidence is plentiful, the methods agree. The serious-injury panel (right) shows where they part.

- With $k = 0$, Waymo's point estimate is "100% reduction" in Los Angeles, Austin and Atlanta (Table 13). The accompanying intervals are valid, but the point estimate carries no information beyond "no events yet".
- For Atlanta, the frequentist interval reaches −87%. The Bayesian posterior gives a median of 88% and a probability of benefit of 0.953, a graded statement that the evidence is favorable but not yet conclusive. At $k = 0$ this probability depends on the prior: it is 0.861 under a flat

prior. A Bayesian result therefore states its prior explicitly, while a frequentist one does not need one.

- For Austin, Waymo's interval just touches zero (−0.2%) and would be labelled "not significant", while the Bayesian probability of benefit is 0.994 (0.975 under a flat prior). The binary significance label obscures how close the evidence is to the threshold.

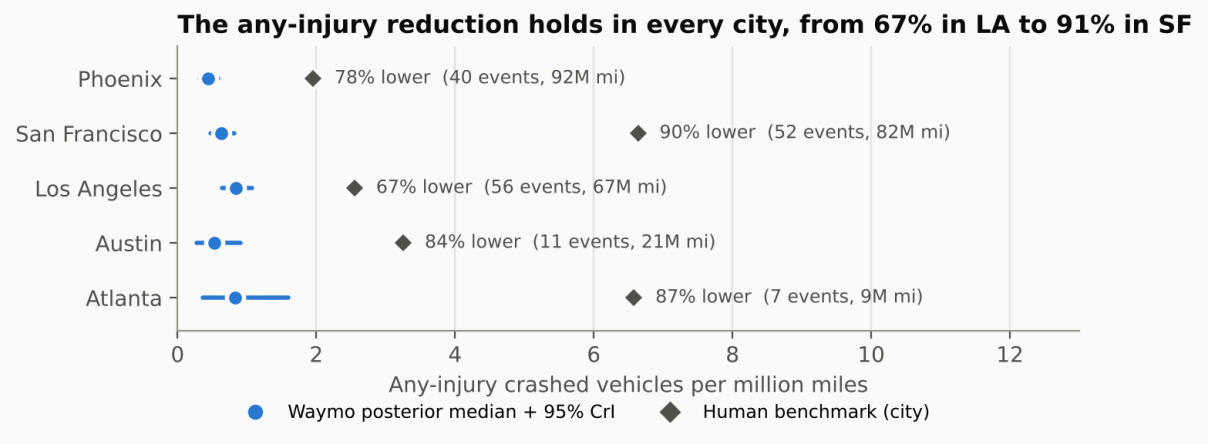


Figure 12. Any-injury posterior by metro area (blue) against the city-specific human benchmark (gray diamond), using this analysis' counts.

Table 14. Any-injury posteriors by metro (this analysis' counts) and Waymo's published city reductions.

Metro	Any-injury *k*	Posterior IPMM (95% CrI)	Benchmark	Reduction, median (2.5%)	Waymo published
Phoenix	40	0.44 (0.32–0.59)	1.95	77.7% (70.0%)	71.1%
San Francisco Bay Area	52	0.63 (0.48–0.82)	6.64	90.5% (87.7%)	89.9%
Los Angeles	56	0.84 (0.64–1.08)	2.55	67.2% (57.9%)	66.2%
Austin	11	0.53 (0.28–0.90)	3.26	83.7% (72.3%)	84.0%
Atlanta	7	0.83 (0.36–1.59)	6.58	87.4% (75.8%)	87.7%

Figure 12 and Table 14 show that the any-injury reduction is large and conclusive in every city. It ranges from 67% in Los Angeles, where Waymo's own rate is highest (0.84 IPMM), to 91% in San Francisco, where the human benchmark is very high (6.64). The independent estimates match Waymo's published city figures within 1 point everywhere except Phoenix (6.6 points), where Waymo counts 52 injury vehicles to our 40. This matches Section 5.2: 12 of Waymo's 20 extra injury vehicles are in Phoenix.

5.6 Miles needed to demonstrate a claim, and power

Table 15. Million miles needed for a 95%-confidence claim, given k observed events (Jeffreys prior). Benchmarks: serious 0.21, airbag 1.61, any injury 3.77 IPMM.

Events observed *k*	Serious, better than human	Serious, ≥ 50% better	Airbag, better than human	Any injury, better than human
0	9.1	18.3	1.2	0.5
1	18.6	37.2	2.4	1.0
2	26.4	52.7	3.4	1.5
3	33.5	67.0	4.4	1.9
5	46.8	93.7	6.1	2.6
10	77.8	155.6	10.1	4.3

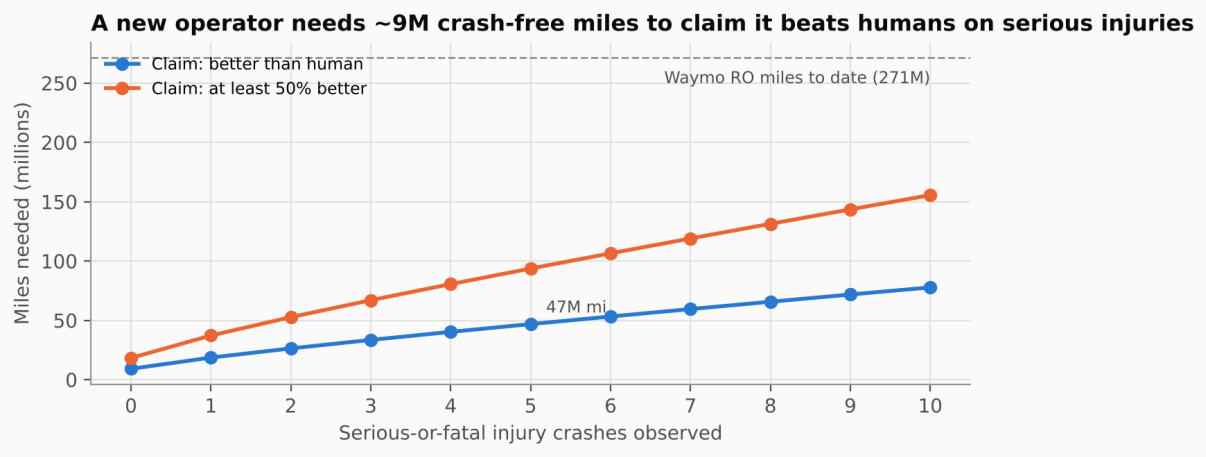


Figure 13. Miles needed against the number of serious-or-fatal crashes observed, for two claims. The dashed line is Waymo's cumulative rider-only mileage.

Figure 13 and Table 15 look backwards: given *k* events, how many miles must have been driven? The answer grows roughly linearly, by 6–9 million miles per serious crash for the "better than human" claim (blue) and twice that for "≥ 50% better" (orange). Waymo's 271 million miles (dashed line) are far beyond the requirement, which is why its serious-injury result is conclusive. Atlanta, at 8.6 million miles with zero events, sits just below the 9.1-million-mile threshold, matching its P(better) of 0.953 in Figure 2.

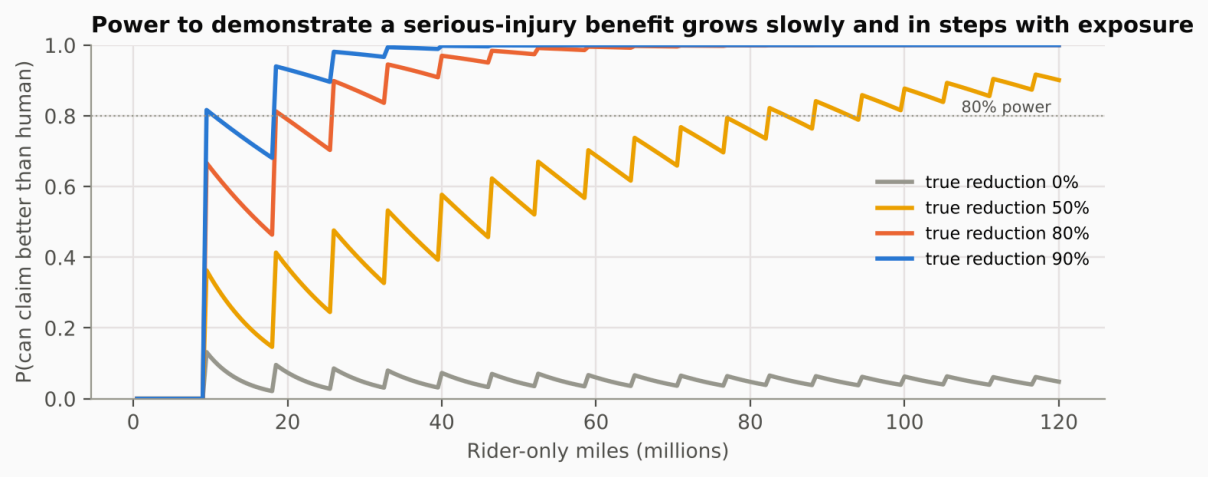


Figure 14. Power: the probability that a fleet can make a 95%-confidence serious-injury claim by a given exposure, for four true reductions.

Table 16. Exposure needed to reach a given power for the serious-injury claim.

True reduction	Miles for 50% power	Miles for 80% power
0% (equal to human)	never	never
50%	33.0 M	82.5 M
80%	9.5 M	18.5 M
90%	9.5 M	9.5 M

Figure 14 looks forwards: for a fleet with a given true safety level, what is the chance it will be able to make the claim? The curves rise in a sawtooth, because the count is discrete. Each time the exposure allows one more event, the claim threshold steps up and power jumps, then decays until the next step. A fleet as good as Waymo (true reduction about 80–90%) reaches 80% power at 9.5–18.5 million miles. A fleet that is "only" 50% better needs 82.5 million miles (Table 16). A fleet equal to humans (gray) almost never qualifies, which is the 5% false-claim rate by design.

Neither Waymo's hub nor its papers provide this operator-facing planning view, although Scanlon et al. (2024) report a related power analysis for benchmark sample size.

5.7 Crash mechanisms and severity: an event-tree view

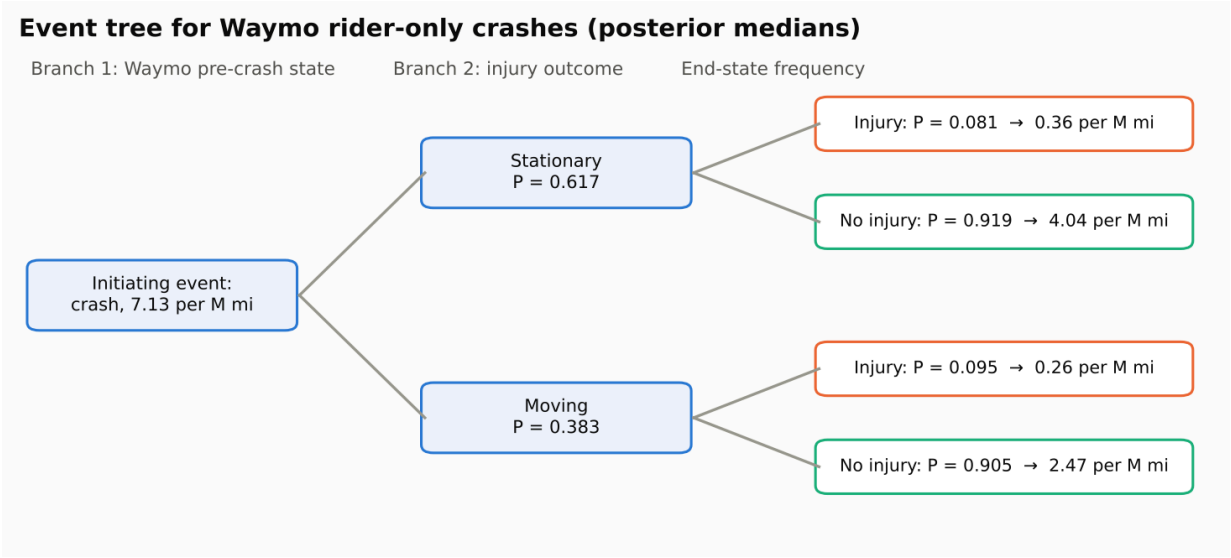


Figure 15. Event tree for Waymo rider-only crashes. The initiating event (any SGO-reported crash) branches on the Waymo vehicle's pre-crash state, then on injury. End states show conditional probability and frequency per million miles (posterior medians).

Table 17. Event-tree quantification. The initiating-event frequency is 1,934 / 271.3 = 7.13 crashed vehicles per million miles.

Branch	Crashes	P(branch)	Injury crashes	P(injury | branch), median (95% CrI)	Injury end-state frequency (IPMM)
Stationary (stopped or parked)	1,194	0.617	96	8.1% (6.6–9.7%)	0.36
Moving	740	0.383	70	9.5% (7.5–11.7%)	0.26
Total	**1,934**	1.000	**166**	—	**0.62**

Figure 15 and Table 17 express the crash record in PRA form. Reading left to right, 7.13 crashed vehicles per million miles enter the tree, and 62% of them occur while the Waymo vehicle is stationary. Because the conditional injury probability is similar on both branches (8.1% and 9.5%), the stationary branch contributes the larger injury end-state frequency (0.36 of 0.62 IPMM). The two injury end states sum to 0.62 IPMM, matching the directly estimated any-injury rate of 0.61 in Table 8, which checks the tree's internal consistency.

In PRA terms, the dominant initiating events are caused by *external agents*, other drivers striking a stopped vehicle, rather than by failures of the ADS itself. Further risk reduction depends on modelling other road users: stopping position, escape paths, defensive behavior.

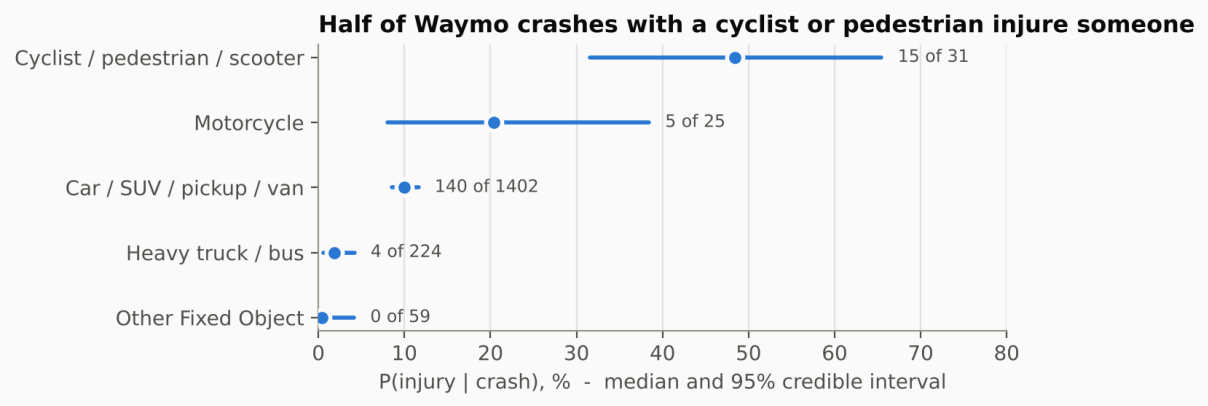


Figure 16. P(injury | crash) by crash partner, median and 95% credible interval.

Table 18. Conditional injury probability by crash partner (Jeffreys Beta posterior). Excluded: Other (161), animal (15), first-responder vehicle (14), pole or tree (3).

Crash partner	Crashes	Injury crashes	P(injury | crash), median (95% CrI)
Cyclist / pedestrian / scooter	31	15	48.4% (31.6–65.5%)
Motorcycle	25	5	20.4% (8.1–38.4%)
Car / SUV / pickup / van	1,402	140	10.0% (8.5–11.6%)
Heavy truck / bus	224	4	1.9% (0.6–4.2%)
Fixed object	59	0	0.4% (0.0–4.2%)

Figure 16 and Table 18 add a second branching dimension, the crash partner. Cyclist, pedestrian and scooter crashes are rare (1.6% of crashes) but injure someone almost half the time, nearly five times the light-vehicle rate. Heavy-truck and bus crashes are common (12%) but rarely injurious; in 86% of them the Waymo vehicle was stationary. The cyclist and pedestrian branch is therefore low-frequency and high-consequence, the classic target for scenario-based testing. This complements Waymo's crash-type rate comparisons (Kusano et al., 2025), which compare frequencies against human benchmarks but do not quantify conditional severity per branch.

Table 19. Conditional injury probability by year (2021–2022 omitted: 15 crashes in total).

Year	Crashes	Injury crashes	P(injury | crash), median (95% CrI)	Stationary share
2023	86	7	8.3% (3.7–15.3%)	50.0%
2024	428	26	6.1% (4.1–8.6%)	66.1%
2025	863	82	9.5% (7.7–11.6%)	64.5%
2026 (Jan–Jun)	542	51	9.4% (7.2–12.1%)	55.7%

Table 19 shows conditional severity broadly stable at 6–10% since 2023. The rise from 2024 to 2025 coincides with expansion into new cities and with the 2025 change in reporting rules. Without

monthly exposure, which Waymo does not publish, this cannot be separated into a rate trend, so it should not be read as a deterioration.

5.8 Human-in-the-loop events

Table 20. Narrative indicators of human involvement, all operating modes, 2025 onward (approximate, keyword-based).

Operator (2025 onward)	Reports	Remote operator in control	Test driver present	Manual mode or takeover	Any human role mentioned
Waymo	1,687	2	76	51	53 (3.1%)
Zoox	105	0	1	47	47 (44.8%)
Avride	82	0	8	3	3 (3.7%)
Tesla	36	5	0	3	22 (61.1%)
May Mobility	26	0	1	4	4 (15.4%)

Table 20 shows that operators differ sharply in the human role recorded in their crash reports. Zoox narratives often describe operator disengagements during supervised testing. Tesla narratives frequently mention safety monitors, and 5 of 36 describe a remote-assistance operator controlling the vehicle at or just before contact, typically a low-speed contact while recovering a stuck vehicle.

Remote assistance moves the human from the driver’s seat to an operations center. Nuclear PRA faced the same shift with actions taken outside the main control room (emergency response, FLEX deployment, remote shutdown panels), where error probability is driven by available time, situational awareness, communication and dependence on automation. We found no explicit model of these actions in Waymo’s published retrospective methodology. As the ratio of vehicles to remote operators grows, they will matter more.

6. Comparison with Waymo’s methodology: gaps and advantages

Table 21 sets the two approaches side by side; Sections 6.1–6.3 interpret the differences.

Table 21. Side-by-side comparison of Waymo's retrospective methodology and this analysis.

Aspect	Waymo (hub and papers)	This analysis	Implication
Crash data	SGO plus internal review: narratives, video, police reports	Public SGO fields only	Waymo's outcome counts are more accurate. This analysis is reproducible by anyone and still matches 99.5% of the crash set (Section 5.2).
Serious-injury definition	Police-confirmed A or K	SGO "Serious" or "Fatality"	Our count of 6 against Waymo's 3 makes this analysis conservative (Table 9).
Benchmarks	Police data and HPMS; under-reporting correction; spatial re-weighting	Waymo's benchmarks used as given	No independent benchmark; this analysis inherits Waymo's benchmark choices.
Rate inference	Exact Poisson and Clopper–Pearson intervals (Nelson, 1970)	Gamma–Poisson posterior (Jeffreys prior)	Same answers when k is large (Figure 11); the Bayesian results remain sensible at $k = 0$ (Table 13).
Output statements	Point estimate, 95% CI, significant or not	Full distribution, percentiles, P(better), exceedance curves	Enables graded statements such as "a 95% probability of at least an 80% reduction" (Figure 8).
Benchmark uncertainty	Poisson sampling in reported intervals; correction factors applied as point estimates (bootstrap intervals for spatial multipliers reported separately in Chen et al., 2024)	Parametric (lognormal CV) sensitivity and a bounding analysis across Waymo's benchmark versions	Headline precision depends on unquantified correction uncertainty (Figure 10, Table 12).
Zero events	Point estimate 100% with a valid interval	Proper posterior, median and P(better), with the prior stated	More informative central estimate; prior-dependent at $k = 0$ (Figure 11, Table 13).
Multiple comparisons	About 1,800 published rows of outcome × location × crash type × benchmark version; no correction (stated in Kusano et al., 2025)	Not corrected either; hierarchical pooling proposed (Section 9)	Isolated "significant" subgroup results should be read with caution.
Prospective planning	Power analysis for benchmark sample size (Scanlon et al., 2024)	Miles-to-demonstrate and operator power curves	Gives operators and regulators claim thresholds in advance (Figures 13–14).
Crash mechanisms	Crash-type rates against matched benchmarks	Event tree with conditional severity per branch	Complementary: Waymo shows *which* crash types

Aspect	Waymo (hub and papers)	This analysis	Implication
			are reduced, this analysis *where severity comes from* (Figures 15–16).
Time variation	Cumulative since 2020; no rate trend	Severity by year; rate trend not possible without monthly miles	Both are limited by the lack of published monthly exposure.
Human-in-the-loop	Outside the scope of the retrospective analyses	Narrative indicators across operators	Identifies an HRA gap (Table 20).
Independence	All authors are Waymo employees (disclosed)	Independent, from public data	Independent reproduction strengthens the credibility of the evidence.

6.1 Gaps in Waymo's approach

1. **Correction-factor uncertainty is not propagated.** The under-reporting correction and spatial adjustment are applied as fixed values. Scanlon et al. (2024) state that the adjustments were considered "point estimates only", and the hub notes that the under-reporting interval is "not straightforward to compute". Figure 10 shows why this matters for any-injury crashes: the spread across Waymo's own constructions (14 points) is more than twice the reported sampling interval (6 points). Not every construction is equally credible, but the corrections that separate them are themselves uncertain.
2. **Binary significance language.** A threshold at zero labels results "significant" or "not", which hides graded evidence. Austin's serious-injury result is "not significant" (−0.2% lower bound) despite a 0.994 probability of benefit (Table 13).
3. **Uninformative zero-event point estimates.** Point estimates of "100% reduction" for cities with no events are mathematically correct but can be misread without the accompanying interval (Figure 11, Table 13).
4. **No correction for multiple comparisons.** With about 1,800 published comparison rows, some will cross a 95% threshold by chance. Kusano et al. (2025) state this explicitly, and report a significant *increase* in one Phoenix subgroup (front-to-rear crashes in which the Waymo vehicle was struck) among their supplemental results.
5. **No operator-facing exposure thresholds.** The hub demonstrates Waymo's own significance. It does not tell other operators, cities or regulators how much exposure a claim requires (Section 5.6).
6. **No treatment of fleet-level dependencies or human operators.** A shared software stack makes crashes across the fleet partially dependent, in the way common-cause failures are in PRA. Remote-assistance actions (Section 5.8) are outside the analysis scope.

6.2 Where Waymo's approach is stronger

1. **Outcome verification.** Police-report confirmation of serious injuries, video review of airbags and narrative review of "Unknown" severities use information this analysis cannot access (Table 7).
2. **Benchmark construction.** Crashed-vehicle-level benchmarks, road-type filtering, under-reporting correction and spatial re-weighting (Scanlon et al., 2024; Chen et al., 2024) reflect years of work. This analysis adopts them rather than replacing them.
3. **Crash-type matching.** Crash-type-specific benchmarks allow comparisons, for example of intersection crashes, that an event tree built from SGO fields alone cannot match.

4. **Transparency.** Publishing the crash list with SGO IDs and every comparison made this independent reconciliation possible.

6.3 Synthesis: combining the strengths

The approaches are complementary, not competing. The strongest evidence comes from Waymo's *verified counts* run through a *Bayesian, uncertainty-propagating model*: the dashed curves in Figure 7. Reporting it with the bounding range across Waymo's own benchmark versions in Figure 10 completes the picture. On that basis, the any-injury evidence supports a reduction of about 82% under Waymo's preferred benchmark, and between roughly 65% and 85% across all four of Waymo's published constructions (Table 12). A safety case would gain from reporting both the headline and the range, and from quantifying the uncertainty of the correction factors that separate them.

7. Discussion: what a PRA lens adds to AV safety cases

Uncertainty belongs in the claim. A point estimate such as "82% fewer injury crashes" is incomplete without its distribution, its exposure and its benchmark assumptions. Figures 6–10 show how much each of these contributes.

Exposure thresholds can be set in advance. The curves in Figures 13–14 can be computed before deployment. They define when each claim becomes testable and give a quantitative trigger for investigation when observed events outrun the curve.

Event trees locate the risk. Splitting crashes by vehicle state and crash partner (Figures 15–16) shows that external-agent initiators dominate frequency and vulnerable-road-user branches dominate severity. This is the AV equivalent of ranking initiating events and accident sequences by risk importance.

Credible evidence needs independent reproduction. Waymo's own framework emphasizes "credibility of the evidence" (Favarò et al., 2023). The 99.5% set agreement in Section 5.2 is exactly that kind of evidence, and it is available to anyone with the public data.

Human reliability does not disappear with the driver. It moves to remote assistance, fleet response and maintenance (Section 5.8).

8. Limitations

- **Data quality.** SGO reports are self-reported, partly redacted, and subject to the 2025 rule change. This analysis cannot reproduce Waymo's manual verification (Table 7).
- **Unit of analysis.** Counts are crashed vehicles, not crashes, matching Waymo's convention.
- **Exposure.** Monthly exposure is not public, so rate trends over time cannot be estimated (Table 19). Waymo's 271.3 million miles include some driving from September 2020 to June 2021, before SGO reporting began; Waymo's list has two crashes from that period.
- **Benchmarks.** The human benchmarks are Waymo's. This analysis quantifies their uncertainty (Figures 9–10) but does not construct independent ones. The equal-weight mixture is a bounding device, not a best estimate: the uncorrected versions are known to be biased low, and weighting by credibility would move the mixture toward the headline.
- **Prior sensitivity.** With large counts the Jeffreys prior has negligible influence. At $k = 0$ or 1 it matters: Atlanta's serious-injury probability of benefit is 0.953 under the Jeffreys prior and 0.861 under a flat prior.

- **Rider-only classification.** Eight 2021 San Francisco and Chandler crashes kept by this analysis are most likely supervised-testing crashes (Section 5.2). Their effect on the aggregate results is negligible (8 of 1,934 vehicles).
- **Homogeneity.** The Poisson model assumes a constant rate. City-level results (Figures 11–12) partly address heterogeneity; a hierarchical model would address it fully.
- **Classification.** Metro assignment uses city names and narrative flags use keywords, so both are approximate.
- **Scope.** Only Waymo has the exposure needed for rate estimates. Results for other operators are descriptive.

9. Future work

1. **A hierarchical Bayesian model** across cities, crash types and software versions (Gamma–Poisson with population variability). This partially pools estimates and is the principled answer to the multiple-comparison problem in Section 6.1.
2. **Independent benchmark uncertainty quantification.** A probabilistic model of under-reporting and spatial adjustment, replacing the equal-weight mixture with evidence-based weights.
3. **Dependency treatment of software releases,** modelling crashes that share a release as a common-cause group, with alpha-factor-style parameters.
4. **An HRA model for remote assistance,** adapting integrated, simulation-informed HRA to remote operators: time available, workload, latency and the ratio of operators to vehicles.
5. **Scenario-level PRA** of the minimal-risk-condition fallback maneuver, with fault trees for redundant sensing and compute.
6. **Bayesian sequential monitoring** to detect rate changes after software releases, once exposure by period is available.

10. Reproducibility

Code:

- `sgo_analysis.py`: data processing and the core model.
- `compare_waymo.py`: reconciliation with Waymo's crash list.
- `comparison_figs.py`: method comparison and figures.

Inputs are the two NHTSA SGO ADS files and Waymo's Safety Impact hub files (crash list and comparison table, through June 2026). Outputs are the cleaned dataset, about 20 result tables (CSV) and 16 figures (PNG). All random draws use fixed seeds, so re-running the scripts reproduces every number in this report.

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